

Customer Segmentation and Prediction Report

This project focuses on customer segmentation and prediction using Python-based analytics techniques. The analysis was performed to understand customer purchasing behavior, identify customer groups, and generate business intelligence insights for future decision-making.

Name	Souvik Bagchi
Tools Used	Python, NumPy, Pandas, Google Colab
Data Cleaning	Python
Visualization	Python
LinkedIn	www.linkedin.com/in/souvik-bagchi-data-analyst
GitHub	https://github.com/Souvik1555
Email	bagchiofficial05@gmail.com

Project Objective

The objective of this project is to perform customer segmentation using machine learning and data analysis techniques to predict customer purchasing behavior and support business growth.

Data Cleaning Process

The dataset was cleaned using Python by handling missing values, removing duplicate records, correcting inconsistent values, and transforming variables for analysis.

Feature Engineering

New features such as customer lifetime value, average monthly spending, and purchase frequency were created to improve customer segmentation analysis.

Customer Segmentation

K-Means clustering was used to divide customers into multiple groups based on purchasing patterns, spending behavior, and transaction frequency.

Dashboard Features

- Customer Cluster Distribution
- Revenue Contribution by Cluster
- Payment Method Analysis
- Purchase Frequency Analysis
- Monthly Spending Pattern
- Top 10 High-Value Customers

Key Insights

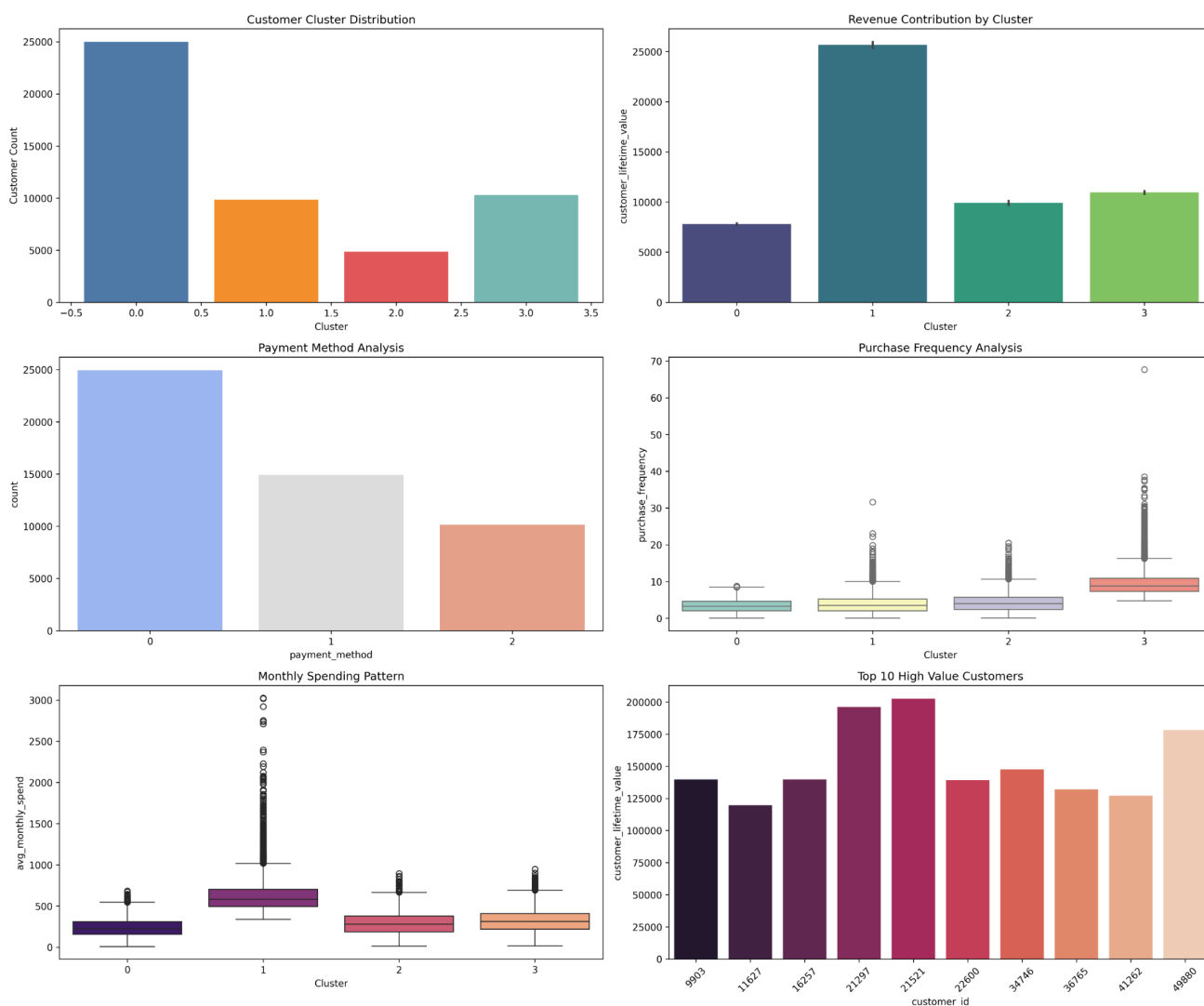
- Certain customer clusters contributed significantly higher revenue.
- High-frequency buyers generated better customer lifetime value.
- Spending patterns varied across customer segments.
- Customer segmentation supports personalized marketing strategies.
- Business intelligence insights help improve customer retention.

Recommendations

- Focus on high-value customer groups.
- Improve loyalty programs for repeat buyers.
- Use predictive insights for personalized promotions.
- Optimize marketing strategies using customer segmentation analysis.

Dashboard Preview

Customer Segmentation & Business Intelligence Dashboard



Conclusion

The Customer Segmentation and Prediction project demonstrates practical data analytics and machine learning using Python, NumPy, Pandas, and visualization libraries. The dashboard provides meaningful business insights that help organizations understand customer behavior, improve decision-making, and optimize business.

strategies.